Vivekanand Education Society's Institute of Technology



Department of Computer Engineering

Group No.: 06

Date :- 09-02-2026

Capstone Project - I Synopsis (2025-26) - Sem VI

Sannidhi: An Agentic Multimodal Framework for Proactive Geriatric Support, Non-Intrusive Activity Recognition, and Privacy-Preserving Health Analytics

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Sannidhi: An Agentic Multimodal Framework for Proactive Geriatric Support, Non-Intrusive Activity Recognition, and Privacy-Preserving Health Analytics

Abstract

The global demographic shift toward a "super-aged" society has created an urgent need for innovative, non-intrusive solutions that support independent aging while ensuring safety. This project introduces Sannidhi, a proactive Agentic AI framework specifically engineered for senior citizens living alone. Unlike traditional reactive systems that rely on manual triggers or constant human supervision, Sannidhi utilizes an autonomous AI agent to initiate supportive interactions based on longitudinal health data. The technical architecture integrates three "hero" domains: Deep Learning (DL) for device-free activity recognition using WiFi Channel State Information (CSI), Natural Language Processing (NLP) for Empathetic Reminiscence Therapy in Hindi and Marathi, and Data Warehousing (DWH) for multi-modal health trend analysis.

To solve the "surveillance feel" that often leads seniors to reject camera-based monitoring, Sannidhi utilizes WiFi ripples to identify human postures through walls and in complete darkness. By processing these signals with a CNN-LSTM architecture, the system aims for >90% detection accuracy for critical events like falls, providing a safety net that requires no wearables. The Agentic AI core acts as the central intelligence, dynamically coordinating all system outputs. When the DWH identifies subtle, negative health trends—such as a 20% decrease in mobility or a shift in vocal sentiment—the Agent autonomously decides on proactive interventions, such as scheduling an empathetic check-in conversation or alerting a remote caregiver via a Streamlit-based dashboard.

By bridging the gap between cutting-edge technology and compassionate care, Sannidhi represents a paradigm shift in geriatric support. The framework prioritizes data sovereignty by implementing Federated Learning and Differential Privacy, ensuring that sensitive biometric and vocal data is processed locally and never leaves the home environment. This software-centric solution not only enhances the physical safety and independence of senior citizens but also addresses the psychological stressors of isolation through personalized, culturally resonant companionship. Ultimately, Sannidhi provides a scalable, research-heavy framework that

transitions elderly care from reactive crisis management to a proactive model that actively preserves vitality and quality of life.

Introduction

Geriatrics Engineering is an interdisciplinary field that applies systems-design principles and advanced engineering technologies to support the aging process. The primary goal is to extend independent living by providing personalized health monitoring and companionship, thereby reducing the strain on formal healthcare systems. However, existing assistive technologies frequently face low adoption rates. Many seniors reject these solutions due to the intrusive "surveillance feel" of camera-based systems or the physical burden of wearables, which can cause skin irritation, require constant charging, and interfere with sleep patterns.

Sannidhi addresses these fundamental limitations by shifting the paradigm from "Passive AI"—which only responds to direct user commands—to **Agentic AI**. While traditional assistants remain dormant until prompted, an Agentic AI possesses the autonomy to learn daily routines, recognize emotional distress through vocal biomarkers, and initiate proactive interventions. This transformation ensures that technology serves as a vigilant but invisible companion, moving beyond simple task execution to active goal-seeking, such as cognitive stimulation and long-term wellness management.

Key Pillars of the Sannidhi Framework:

- **Non-Intrusive Sensing (WiFi CSI):** By utilizing low-cost WiFi signals (Channel State Information) instead of cameras, Sannidhi identifies human activities through walls and in complete darkness. This preserves the user's dignity and privacy while maintaining 24/7 safety monitoring.
- **Proactive Goal-Driven AI:** The system acts autonomously to achieve specific outcomes, such as ensuring medication adherence, encouraging physical movement, or providing empathetic cognitive support through reminiscence therapy.
- **Preservation of Autonomy:** The framework allows seniors to "age-in-place" safely, maintaining their independence by automating routine tasks and detecting critical incidents—like falls—without requiring the user to wear a device or click a button.
- **Localized Privacy Architecture:** To combat data security concerns, Sannidhi implements **Federated Learning** and **Differential Privacy**, ensuring that sensitive

biometric and vocal data are processed on local edge devices rather than in the cloud.

By bridging the gap between cutting-edge signal processing and compassionate, localized AI, Sannidhi not only enhances the physical safety of the elderly but also addresses the silent psychological stressors of isolation. This approach ensures that the "super-aged" society can benefit from high-tech intervention without sacrificing the comfort and privacy of their own homes.

Problem Statement

Senior citizens living alone are vulnerable to three primary threats:

- **Safety and Physical Vulnerability:** Falls are a leading cause of injury for seniors, yet continuous monitoring is often rejected because traditional systems are perceived as privacy-intrusive or require constant user interaction like alert buttons. Changes in respiratory and heart rates are also key predictors of cardiac arrest and falls, but these vital signs are difficult to capture unobtrusively in uncontrolled home environments.
- **Psychological and Social Isolation:** Loneliness is strongly correlated with accelerated cognitive decline, yet many existing technologies are transactional and lack the empathetic reasoning needed to foster meaningful, human-like relationships. Traditional AI follows predefined instructions rather than adapting to an individual's communication style or proactively engaging in companionship.
- **Linguistic and Technical Barriers:** A significant digital literacy divide exists, largely because advanced AI tools are English-centric, leaving a critical gap for empathetic, high-accuracy conversational AI in Indian languages like Marathi and Hindi. Linguistic localization is a primary determinant of accessibility for the growing geriatric population in multilingual societies.

Proposed Solution

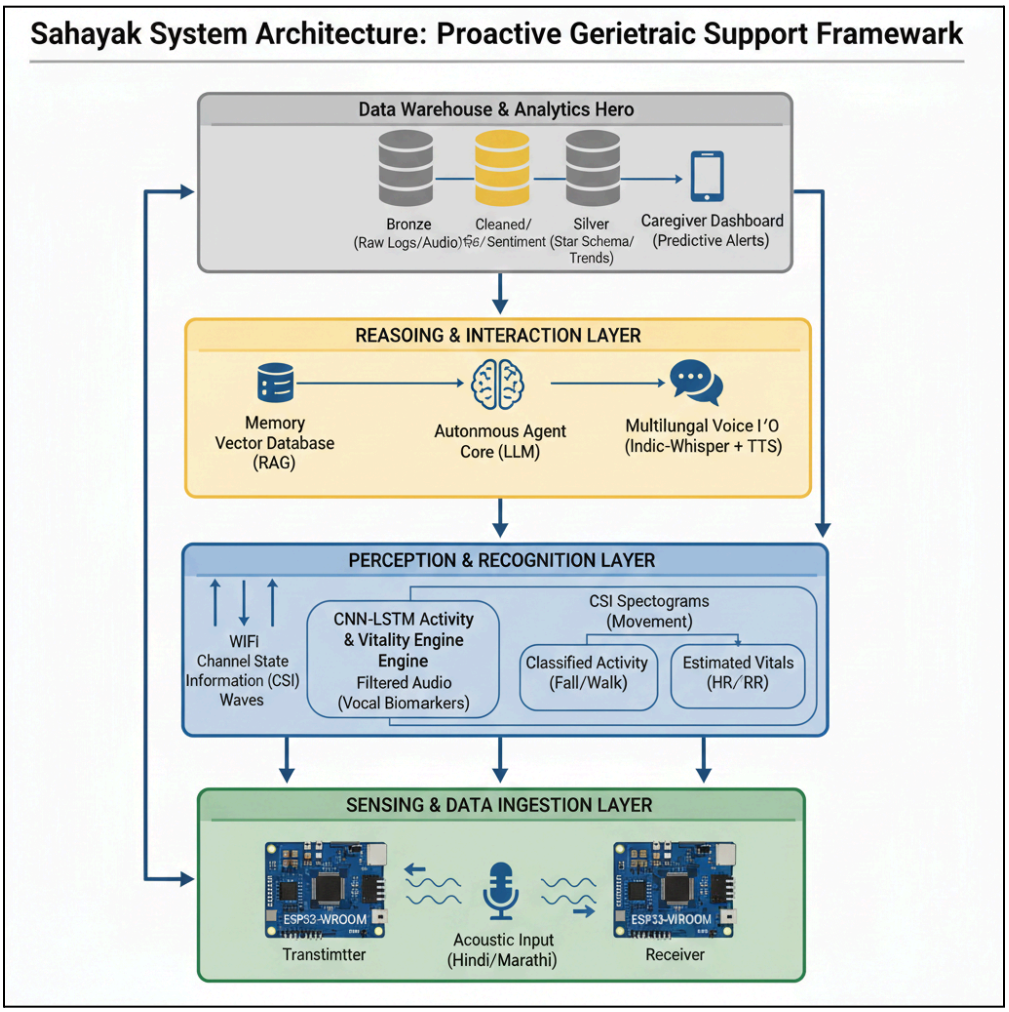
Sannidhi is a modular, software-centric system designed for comprehensive geriatric care:

- **Routine Guardian (Deep Learning):** This component uses WiFi CSI sensing to track movement and vital signs unobtrusively. By capturing detailed spatial and temporal information about signal interactions with the environment, it detects falls and monitors

cardiorespiratory activity without cameras or wearables.

- **Reminiscence Companion (Agentic AI):** Powered by LLMs, this agent conducts Reminiscence Therapy (RT) by evoking past experiences through conversations, which stimulates autobiographical episodic memory. It uses Retrieval-Augmented Generation (RAG) to remember personal history and can autonomously tailor its communication style to provide cognitive support and emotional engagement.
- **Health Historian (Data Warehouse):** The system implements a **Medallion Architecture** (Bronze, Silver, Gold layers) to progressively improve health data quality. It organizes diverse data—from activity logs to sentiment scores—into project-specific schemas optimized for clinical trend analysis and predictive analytics.

Methodology / Block Diagram



1. **Non-Intrusive Sensing (Deep Learning):**
    - The sensing layer captures WiFi Channel State Information (CSI), which offers a comprehensive representation of signal amplitude and phase across multiple subcarriers. This allows the system to recognize fine-grained movements even behind walls by analyzing multipath distortions caused by body motions.
    - Pre-processing involves noise removal via Kalman filters or Discrete Wavelet Transforms, followed by a **CNN-LSTM** or **GRU** deep learning architecture for time-series feature extraction. These models establish robust relationships between specific CSI patterns and activities like lying down, falling, or sitting.
2. **Agentic Reminiscence & Sentiment Analysis (NLP):**
    - The interaction layer employs LLM-guided conversation to steering discourse through autobiography interviewing protocols and empathetic engagement. It integrates RAG to access proprietary user memories and uses medical NLP to analyze clinical notes and sentiment, distinguishing between current and historical health indicators.
    - A memory graph is designed to drive memory extrapolation, enhancing the agent's goal navigation capabilities to lead meaningful therapy sessions.
3. **Data Warehousing & Analytics (DWH):**
    - Sannidhi follows a tiered Medallion Architecture: the **Bronze layer** acts as an append-only raw zone for original IoT logs; the **Silver layer** cleanses and standardizes this data using rules based on modification timestamps; and the **Gold layer** refine this into aggregated datasets ready for high-value insights.
    - The architecture enables high-performance pipelines that transform chaotic raw data into a strategic asset for identifying at-risk populations and optimizing care delivery.
4. **Privacy-Preserving Architecture:**
    - Sensitive information is protected through **Federated Learning (FL)**, where

models are trained on decentralized local data sources, and only model updates (like gradients) are shared centrally.

- To further shield patient data, **Differential Privacy (DP)** introduces random noise into updates, making it difficult for attackers to determine specific data sample participation in the training process.

Hardware, Software and Tools Requirements

- **Hardware:**
  - **ESP32-WROOM Modules:** These low-cost commercial WiFi devices act as the sensing platform, capturing fine-grained CSI amplitude and phase data.
  - **Local Host (Laptop/Raspberry Pi):** Serves as the edge node for running local LLM quantization and deep learning inference, ensuring raw data remains on-site.
  - **Standard Microphone & Speakers:** Essential for high-quality audio capture and synthesized voice interaction during therapy sessions.
- **Software:**
  - **Deep Learning:** Frameworks like TensorFlow and Keras are used to implement CNN-LSTM and GRU-based activity recognition models.
  - **NLP & Agentic Logic:** LangChain and RAG architectures manage context and retrieval of episodic memories, while IndicWhisper provides automated speech recognition for multilingual support.
  - **DWH:** PostgreSQL or Snowflake-style Medallion architectures organize structured and unstructured health data.
  - **Privacy:** PRiMIA or similar frameworks for end-to-end privacy-preserving deep learning using DP-SGD.

Proposed Evaluation Measures

To ensure Sannidhi meets the rigorous standards of a Computer Engineering Capstone, the evaluation is divided into four technical "Hero" metrics that validate the system's performance, empathy, and data integrity:

- **Activity Recognition (Deep Learning):** We will use CNN-LSTM and GRU models, evaluated via confusion matrices to calculate F1-score, Recall, and Specificity. Our primary target is achieving >90% accuracy for critical "Sudden Falls" detection in both Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) environments. The accuracy of vital signs monitoring, such as respiratory rates, will be benchmarked by comparing WiFi CSI-derived data to commercial wearable sensors, using the Mean Absolute Error (MAE) as the metric. We will also employ the Receiver Operating Characteristic (ROC) curve and Area Under the Curve (AUC) for a comprehensive evaluation of model discrimination capability.
- **Conversational Empathy (Agentic AI):** The quality of Reminiscence Therapy (RT) delivered by the AI will be measured using the Chatbot Compassion Quotient (CCQ) and "Event Coverage" checks to ensure high story reconstruction fidelity. We will employ IndicWhisper to evaluate the Word Error Rate (WER) for Marathi and Hindi speech-to-text. The system's capacity for proactive engagement will be tracked using the "Proactivity Index," which compares Agent-initiated versus user-initiated interactions. Furthermore, user interviews and thematic analysis of session transcripts will be used to qualitatively assess the perceived empathy and therapeutic value.
- **Privacy & Utility (Security):** We plan to utilize Differential Privacy (DP) and Federated Learning (FL) to rigorously evaluate the "Privacy Budget" and determine the optimal balance between data obfuscation and model utility. Evaluation will include measuring the Accuracy Loss caused by DP-SGD noise to ensure the reliability of critical functions like fall detection is not compromised. Furthermore, "Latency at the Edge" will be measured to guarantee the system can perform near real-time local inference for responsive action. We will also perform a robust attack-resistance test against common inference attacks, such as Membership Inference Attacks (MIA), to validate the security measures.

Data Warehouse Integrity & Predictive Readiness (DWH): The Medallion Architecture will be

evaluated based on data latency (time taken for a "Bronze" raw log to become a "Gold" actionable insight). The Gold Layer Star Schema will be stress-tested for query performance to ensure the Caregiver Dashboard can generate longitudinal health trend graphs—such as "3-Month Mobility Decline"—instantly. Success is defined by the DWH's ability to generate "Early Warning" flags that correlate sentiment drops with activity decreases, providing a "Predictive Accuracy" score for identifying potential depressive episodes.

Conclusion

Sannidhi aims to prove that high-end engineering—integrating Agentic AI, Deep Learning, and Data Warehousing—can be synthesized into a solution that is profoundly human-centric and accessible. By prioritizing non-intrusive sensing through WiFi CSI and providing culturally resonant linguistic support in Hindi and Marathi, the project directly addresses the core psychological and physical challenges of the burgeoning elderly population in India. This framework shifts the paradigm of senior care from reactive crisis management, which often arrives too late, to a proactive, empathetic support system that monitors safety without the "surveillance feel" of cameras.

Ultimately, this capstone project establishes a scalable blueprint for the future of "Geriatrics Engineering" by demonstrating how complex, decentralized data can be managed securely. By implementing Federated Learning and Differential Privacy, Sannidhi ensures that a senior's most private daily routines never leave their home, setting a new standard for trust in the smart home ecosystem. The successful deployment of this framework will enable older adults to maintain their independence and dignity for longer, while providing their families with data-driven peace of mind through a robust, technically sophisticated, and empathetic digital companion.

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